

CUBIC MARKED-ROOT COVERS: NORMAL FORMS, EQUIVALENCE, AND STABLE UNIQUENESS

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ABSTRACT. The recent counterexample to the Jacobian conjecture is governed by a binary cubic together with a marked simple root. We study two classification questions suggested by that description.

First, we classify a normalized determinant-neutral family in which the inverse cubic has the form

$$A(c)T^3 + B(c)T^2 + bT - 2a.$$

Polynomiality is equivalent to a finite set of jet conditions at a distinguished simple zero of A . When A and B are coprime, the projective root-incidence completion is a smooth finite flat cubic cover, and we determine its boundary, discriminant, nonproperness set, omitted locus, and fiber stratification. Without a squarefreeness or coprimality hypothesis, we classify the resulting maps up to arbitrary polynomial left–right equivalence: the invariant is an affine orbit of the divisor of A , its marked simple point, and the residue class of B modulo A . The coordinate frame is governed by a universal root–slope identity: the root-forgetting Jacobian is $P'(t)/2$, while the affine chart contributes its reciprocal. This gives generic S_3 monodromy and a product description of the principal discriminant complement.

Second, we consider the larger binary multiplication-incidence construction obtained from $\mathbb{P}(V_a) \times \mathbb{P}(V_b) \rightarrow \mathbb{P}(V_{a+b})$. After removing the resultant and the pullback of any tangent but nonosculating hyperplane, the resulting open is stably affine space only in the cubic case $\{a, b\} = \{1, 2\}$. The obstruction combines the public class-group reduction to adjacent degrees with a codimension-two Hodge–Deligne deficit.

Working manuscript. The computer checks described in section 7 are exact, but they do not replace independent review of the geometric arguments.

1. INTRODUCTION

A polynomial map $F: \mathbb{C}^n \rightarrow \mathbb{C}^n$ is a *Keller map* if its Jacobian determinant is a nonzero constant. The Jacobian conjecture asserted that every Keller map is a polynomial automorphism. The explicit three-dimensional counterexample disproved the conjecture in dimensions at least three.

Discovery attribution. We credit Akhil Mathew as the primary human source of the problem: Mathew prompted Levent Alpöge, Alpöge prompted

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Fable, Fable produced the example, and Alpöge announced it publicly [Ula, Alp].

The feature relevant here is not the original coordinate formula but its root-incidence geometry. The public projective formulation due to Jiang [Jia] regards a source point as a binary cubic together with a marked simple root; the map forgets the mark. Lou gave a resultant-normalized affine chart [Lou], and Litt described the source through affine-line bundles and pencils [Lit]. Speyer’s discussion [Spe], Tao’s exposition [Tao], and Naskręcki’s audit [Nas] contain further covering, boundary, and orbit interpretations.

Our first question is what remains after fixing the determinant-neutral coordinates but allowing the cubic coefficients to vary with one target coordinate. This produces the normalized A, B family of section 3. Pole cancellation is rigid enough to classify every member of the family, while flexible enough to retain arbitrarily complicated one-variable data. The main results are the global description in the coprime case, theorem 3.4, and the unrestricted left–right classification, theorem 4.1.

Our second question is whether the cubic marked-root construction is one member of a higher binary multiplication family with affine-space source. The class-group obstruction in this direction was already observed publicly by Skozi and Shubhodip Mondal; Mondal also reported the first $(a, b) = (2, 3)$ point-count failure in the Secret Blogging Seminar discussion [Spe]. We take that reduction as the starting point. Theorem 5.1 proves that every adjacent higher pair has a universal codimension-two Hodge–Deligne deficit. Thus the cubic is the unique stably affine case throughout the two-block construction.

Novelty boundary. We do not claim the counterexample, its marked-root interpretation, the resultant chart, the basic $3/1/0$ fiber count, or the initial class-group obstruction. Our candidate contributions are:

- (i) the complete pole-cancellation and coprime global-geometry theorem for the normalized A, B family;
- (ii) the all-multiplicity left–right classification in that family; and
- (iii) the codimension-two argument proving stable uniqueness in the full two-block incidence family.

For orientation we also record the root–slope identity, generic monodromy, and discriminant-complement geometry of the normalized frame. These organize the classification but are not used to enlarge the novelty claim.

The first two results concern a deliberately restricted normal form; they do not classify arbitrary degree-three Keller maps. The last result concerns a specific binary multiplication construction; it is not a universality theorem for counterexamples.

Conventions. We work over $\mathbb{C}$. Varieties are reduced and separated. We write $\mathbb{L} = [\mathbb{A}^1]$ in the Grothendieck ring $K_0(\text{Var}_{\mathbb{C}})$, and $F_d K_0$ for the subgroup generated by varieties of dimension at most d . Compactly supported Hodge–Deligne polynomials are denoted by $E_c(-; u, v)$.

2. THE MARKED-ROOT COORDINATE FRAME

Let $A, B \in \mathbb{C}[c]$, with $A \neq 0$, and initially work on the chart $x \neq 0$. Put

$$t = y + \frac{1}{x}, \quad r = \frac{2}{x}, \quad c = w(x, y) - x^3 z, \quad (2.1)$$

where $w \in \mathbb{C}[x, y]$. Define

$$b = r - 3A(c)t^2 - 2B(c)t, \quad 2a = \frac{2t}{x} - 2A(c)t^3 - B(c)t^2. \quad (2.2)$$

The associated inverse polynomial is

$$P_{a,b,c}(T) = A(c)T^3 + B(c)T^2 + bT - 2a. \quad (2.3)$$

Whenever (a, b, c) extends polynomially across $x = 0$, we denote the resulting map by $F: \mathbb{A}_{x,y,z}^3 \rightarrow \mathbb{A}_{a,b,c}^3$.

The coordinates are designed so that

$$P_{F(x,y,z)}(t) = 0, \quad P'_{F(x,y,z)}(t) = \frac{2}{x} \quad (2.4)$$

on $x \neq 0$. Consequently each finite simple root t reconstructs a source point by

$$x = \frac{2}{P'(t)}, \quad y = t - \frac{P'(t)}{2}, \quad z = \frac{w(x, y) - c}{x^3}. \quad (2.5)$$

Terms in the inverse polynomial of degrees at most one in T introduce no new geometry here: a term $C(c)T + D(c)$ is removed by a triangular target automorphism. Likewise, replacing w by $w + x^3 u(x, y)$ is the source shear $z \mapsto z - u(x, y)$.

3. CLASSIFICATION AND GEOMETRY OF THE A, B FAMILY

3.1. Pole cancellation.

Theorem 3.1 (Pole-cancellation theorem). *Let $A \neq 0$. The rational expressions in (2.2) are polynomials if and only if there is $\alpha \in \mathbb{C}$ such that*

$$A(\alpha) = 0, \quad A'(\alpha) \neq 0, \quad B(\alpha) = -2, \quad (3.1)$$

and, modulo $x^3 \mathbb{C}[x, y]$,

$$w(x, y) \equiv \alpha + \frac{2}{A'(\alpha)}x + x^2 \left(-\frac{3}{A'(\alpha)}y - \frac{2A''(\alpha)}{A'(\alpha)^3} - \frac{B'(\alpha)}{A'(\alpha)^2} \right). \quad (3.2)$$

The x^3 -remainder is removed by a source shear.

Proof. Write $q = 1 + xy$. Clearing the possible poles gives

$$x^2b = 2x - 3A(c)q^2 - 2xB(c)q, \quad (3.3)$$

$$2x^3a = 2xq - 2A(c)q^3 - xB(c)q^2. \quad (3.4)$$

Polynomiality first gives $A(w(0, y)) = 0$. Since A has finitely many roots, $w(0, y)$ is constant; call it α . Write

$$w = \alpha + xp(y) + x^2s(y) + O(x^3),$$

and abbreviate $A_i = A^{(i)}(\alpha)$, $B_i = B^{(i)}(\alpha)$. The coefficients of x in (3.3) and (3.4) are respectively

$$2 - 3A_1p - 2B_0, \quad 2 - 2A_1p - B_0.$$

Their vanishing gives $A_1p = 2$ and $B_0 = -2$. In particular $A_1 \neq 0$ and $p = 2/A_1$ is constant. The coefficient of x^2 in (3.4) then gives

$$s(y) = -\frac{3}{A_1}y - \frac{2A_2}{A_1^3} - \frac{B_1}{A_1^2}.$$

These conditions are also sufficient: substitution makes the right sides of (3.3) and (3.4) divisible by x^2 and x^3 , respectively. Finally, an $x^3u(x, y)$ term in w is absorbed by the stated source shear. $\square$

Definition 3.2. An *admissible triple* is

$$(A, B, \alpha), \quad A, B \in \mathbb{C}[c], \quad A \neq 0,$$

satisfying (3.1). We use the representative $w = w_{A,B,\alpha}$ displayed in (3.2), with no x^3 remainder, and write $F_{A,B,\alpha}$ for the resulting polynomial map.

Proposition 3.3. *Every admissible triple defines a Keller map with*

$$\det DF_{A,B,\alpha} = -2.$$

Its generic degree is three. The source plane $x = 0$ maps isomorphically onto the target plane $c = \alpha$.

Proof. On $x \neq 0$, direct differentiation gives

$$\det \frac{\partial(a, b, c)}{\partial(t, r, c)} = \frac{r}{2}, \quad \det \frac{\partial(t, r, c)}{\partial(x, y, z)} = -2x.$$

Since $r = 2/x$, their product is -2 , and polynomiality extends the identity to all of $\mathbb{A}^3$.

For a general target point, (2.3) has three finite simple roots, and (2.5) gives three distinct source points. Thus the generic degree is three.

For completeness, restriction of the canonical representative to $x = 0$ gives

$$b = y - \frac{B'(\alpha)}{A'(\alpha)}$$

and

$$a = A'(\alpha)z + 4y^2 + \left(\frac{5B'(\alpha)}{2A'(\alpha)} + \frac{6A''(\alpha)}{A'(\alpha)^2} \right) y + \kappa,$$

where $\kappa \in \mathbb{C}$ depends only on the first three derivatives of A and the first two derivatives of B at α . This is a triangular polynomial isomorphism from $\mathbb{A}_{y,z}^2$ to $\mathbb{A}_{a,b}^2$. $\square$

3.2. The coprime completion. Assume for the remainder of this subsection that

$$\gcd(A, B) = 1. \quad (3.5)$$

Let

$$\tilde{X} = V\left(A(c)T^3 + B(c)ST^2 + bS^2T - 2aS^3\right) \subset \mathbb{A}_{a,b,c}^3 \times \mathbb{P}_{[S:T]}^1. \quad (3.6)$$

For a distinct zero β of A , let

$$E_\beta = \{c = \beta, [S : T] = [0 : 1]\} \simeq \mathbb{A}^2.$$

Let R be the repeated-root divisor in $\tilde{X}$.

Theorem 3.4 (Global geometry of the coprime family). *For an admissible triple satisfying (3.5):*

- (i) *the total space $\tilde{X}$ is smooth, and the projection $\pi: \tilde{X} \rightarrow \mathbb{A}_{a,b,c}^3$ is finite flat of degree three;*
- (ii) *there is an isomorphism*

$$\mathbb{A}_{x,y,z}^3 \simeq \tilde{X} \setminus \left(R \cup \bigcup_{\substack{\beta \in Z(A) \\ \beta \neq \alpha}} E_\beta \right); \quad (3.7)$$

- (iii) *the branch discriminant is*

$$\mathcal{D} = B(c)^2b^2 - 4A(c)b^3 + 8aB(c)^3 - 36aA(c)B(c)b - 108a^2A(c)^2; \quad (3.8)$$

- (iv) *the nonproperness set is*

$$S_F = V(\mathcal{D}) \cup \bigcup_{\substack{\beta \in Z(A) \\ \beta \neq \alpha}} V(c - \beta); \quad (3.9)$$

- (v) *if*

$$T_3 = V\left(3A(c)b - B(c)^2, 54A(c)^2a + B(c)^3\right)$$

and

$$C_\beta = V\left(c - \beta, b^2 + 8aB(\beta)\right),$$

then

$$\mathbb{A}^3 \setminus F(\mathbb{A}^3) = T_3 \cup \bigcup_{\beta \neq \alpha} C_\beta = \text{Sing}(S_F)_{\text{red}}. \quad (3.10)$$

Proof. The equation in (3.6) is a nonzero binary cubic on every geometric target fiber: at a common zero of A and B this could fail, but (3.5) excludes such a point. It therefore defines a relative effective Cartier divisor of degree three in the trivial $\mathbb{P}^1$ -bundle. The divisor is proper with zero-dimensional

fibers and hence finite; the constant relative Hilbert polynomial makes it flat of degree three.

The total space is smooth. On $S \neq 0$, the partial derivative with respect to a is $-2S^3$. If $S = 0$, then $T \neq 0$, $A(c) = 0$, and $B(c) \neq 0$; the partial derivative with respect to S is $B(c)T^2$.

On the chart of finite simple roots, (2.5) identifies the incidence with the source. The root at infinity over $c = \alpha$ is also simple because $B(\alpha) = -2$, and proposition 3.3 identifies it with the plane $x = 0$. At every other zero β of A , the simple infinity section E_β has no affine source point. Removing these sections and the repeated-root divisor proves (3.7).

The discriminant of (2.3) is (3.8). On the chosen-root incidence chart, put

$$r = P'(t), \quad b = r - 3A(c)t^2 - 2B(c)t, \quad a = -A(c)t^3 - \frac{1}{2}B(c)t^2 + \frac{1}{2}rt.$$

Then

$$\pi^*\mathcal{D} = r^2 \left((3A(c)t + B(c))^2 - 4A(c)r \right). \quad (3.11)$$

Thus $R = \{r = 0\} \simeq \mathbb{A}_{t,c}^2$, and its image is the branch hypersurface $V(\mathcal{D})$. Each deleted E_β maps isomorphically to the plane $c = \beta$, giving (3.9).

A cubic on the branch hypersurface has one double and one simple root away from T_3 ; the double root lies in R , while the simple root remains. At T_3 all three roots coincide and none remains in (3.7). On $c = \beta \neq \alpha$, the infinity root is deleted. The remaining quadratic has a double root precisely on

$$b^2 + 8aB(\beta) = 0,$$

because

$$\mathcal{D}|_{c=\beta} = B(\beta)^2(b^2 + 8aB(\beta)).$$

This proves the first equality in (3.10).

Where $A(c) \neq 0$, depressing the cubic identifies its branch equation transversely with a cusp; its singular locus is T_3 . Along $c = \beta$, the branch hypersurface is smooth because

$$\left. \frac{\partial \mathcal{D}}{\partial a} \right|_{A=0} = 8B(\beta)^3 \neq 0,$$

and it meets the plane $c = \beta$ transversely exactly along C_β . The reduced singular locus of the union (3.9) is therefore the right side of (3.10). $\square$

Corollary 3.5 (Fiber table). *The affine fiber size is three off S_F , one on $V(\mathcal{D}) \setminus (T_3 \cup \bigcup C_\beta)$, two on a deleted plane away from $V(\mathcal{D})$, and zero on T_3 and on every C_β .*

Remark 3.6. The generic local behavior of an arbitrary cubic normalization is more finely divided than the slogan “ramified, unramified, or boundary.” Relative to the finite completion, the generic divisor types are: three unramified sheets with one deleted; three unramified sheets with two deleted; a $2 + 1$ ramified fiber with the ramified point deleted; and the excluded

generic three-cycle case. We do not use this local classification in the proofs below.

4. POLYNOMIAL LEFT–RIGHT EQUIVALENCE

We now drop the coprimality and squarefreeness assumptions. Two maps $F_i: \mathbb{A}^3 \rightarrow \mathbb{A}^3$ are *left–right equivalent* if there are polynomial automorphisms Φ of the source and Ψ of the target such that

$$F_2 \circ \Phi = \Psi \circ F_1.$$

Theorem 4.1 (All-multiplicity equivalence criterion). *Let (A_i, B_i, α_i) be admissible triples. Then F_{A_1, B_1, α_1} and F_{A_2, B_2, α_2} are polynomially left–right equivalent if and only if there are an affine map $\phi(c) = uc + v$, $u \neq 0$, and a scalar $\kappa \in \mathbb{C}^*$ such that*

$$A_2(\phi(c)) = \kappa A_1(c), \quad (4.1)$$

$$B_2(\phi(c)) \equiv B_1(c) \pmod{A_1(c)}, \quad (4.2)$$

$$\phi(\alpha_1) = \alpha_2. \quad (4.3)$$

No squarefreeness hypothesis is required.

When A_1 has at least two distinct roots, every equivalence induces the root-affine data above. When A_1 is linear, all members lie in one equivalence class; we do not assert that the construction below describes the full self-equivalence group of that class.

4.1. The reduced nonproperness divisor. Let

$$\Delta = B^2b^2 - 4Ab^3 + 8aB^3 - 36aABb - 108a^2A^2.$$

Regard Δ as a polynomial in a, b over $\mathbb{C}[c]$. Let $g(c)$ be the gcd of its coefficients and put

$$\Delta_{\text{prim}} = \Delta/g(c).$$

At a zero β of A , if $m_\beta = \text{ord}_\beta A$ and $n_\beta = \text{ord}_\beta B$, then

$$\text{ord}_\beta g = \min(m_\beta, 2n_\beta).$$

Lemma 4.2. *The polynomial Δ_{prim} is irreducible in $\mathbb{C}[a, b, c]$.*

Proof. Over $\mathbb{C}(b, c)$, the discriminant is quadratic in a , and the discriminant of that quadratic is

$$-64(3A(c)b - B(c)^2)^3.$$

The factor $3Ab - B^2$ has a simple zero as a rational function of b , so it is not a square in $\mathbb{C}(b, c)$. The quadratic is irreducible over that field. Dividing by coefficient content and applying Gauss’s lemma proves the assertion. $\square$

Lemma 4.3. *As a reduced set,*

$$S_F = V(\Delta_{\text{prim}}) \cup \bigcup_{\substack{\beta \in Z(A) \\ \beta \neq \alpha}} V(c - \beta). \quad (4.4)$$

The first component is the unique nonplane irreducible component.

Proof. At a general point of $V(\Delta_{\text{prim}})$ with $A(c) \neq 0$, the inverse cubic has one double finite root. Splitting that root makes $P'(t) \rightarrow 0$, so $x = 2/P'(t) \rightarrow \infty$. Thus a dense open subset of the irreducible hypersurface is nonproper, and closedness of the nonproperness set gives the entire component.

If $A(\beta) = 0$ with $\beta \neq \alpha$, then the limiting degree drops by one when $B(\beta) \neq 0$, and by two when $B(\beta) = 0$. The lost roots tend to infinity. Their source reconstructions cannot remain bounded: otherwise $t \rightarrow \infty$ would force $x \rightarrow 0$, and $c = w(x, y) - x^3 z$ would tend to the distinguished value α , contrary to $\beta \neq \alpha$. Hence every indicated plane is a nonproperness component.

Conversely, away from the displayed set, all roots are finite and simple, and (2.5) is regular. At $c = \alpha$, the projective infinity root is simple and retained by the plane $x = 0$, so the same conclusion holds.

Finally, over $\mathbb{C}(c)$ the primitive discriminant is a cusp in affine coordinates, hence singular. The remaining components in (4.4) are smooth planes. This proves uniqueness. $\square$

4.2. Necessity.

Proof of necessity in theorem 4.1. Suppose

$$F_2 \circ \Phi = \Psi \circ F_1.$$

Polynomial automorphisms preserve nonproperness sets. Assume first that A_1 has at least two distinct roots. By lemma 4.3, a plane component cannot map to the singular primitive-discriminant component. The pullback of the prime ideal of a target plane is therefore the prime ideal of a source plane, giving

$$c_2 \circ \Psi = uc + v = \phi(c), \quad u \neq 0. \quad (4.5)$$

Put $K = \mathbb{C}(c)$,

$$\tilde{A}_2 = A_2(\phi(c)), \quad \tilde{B}_2 = B_2(\phi(c)),$$

and define the cubic covariants

$$p_i = 3A_i b_i - B_i^2, \quad Q_i = 2B_i^3 - 9A_i B_i b_i - 54A_i^2 a_i.$$

The affine change $(a_i, b_i) \leftrightarrow (p_i, Q_i)$ has determinant $162A_i^3$ over K , and

$$Q_i^2 + 4p_i^3 = -27A_i^2 \Delta_i. \quad (4.6)$$

Thus the generic nonplane component is the monomial cusp $Q^2 + 4p^3 = 0$. By the classification of plane automorphisms preserving a monomial cusp [BS15, Theorem 2(ii)], applied over the perfect field K , there is $\rho \in K^*$ such that

$$p_2 \circ \Psi = \rho^2 p_1, \quad Q_2 \circ \Psi = \rho^3 Q_1. \quad (4.7)$$

Write $\hat{a} = \Psi^*(a_2)$ and $\hat{b} = \Psi^*(b_2)$. Comparing coefficients in (4.7) gives

$$\hat{b} = \lambda_b(c)b + f(c), \quad \lambda_b = \frac{\rho^2 A_1}{\tilde{A}_2},$$

and

$$\hat{a} = \lambda_a(c)a + g(c)b + h(c), \quad \lambda_a = \frac{\rho^3 A_1^2}{\tilde{A}_2^2}.$$

Polynomiality puts the displayed coefficients in $\mathbb{C}[c]$. Since

$$\det D\Psi = u\lambda_a\lambda_b \in \mathbb{C}^*,$$

both λ_a and λ_b are constants. It follows that $\rho = \lambda_b^2/\lambda_a \in \mathbb{C}^*$ and that $\tilde{A}_2/A_1 = \kappa \in \mathbb{C}^*$, proving (4.1) with multiplicities.

The plane components are indexed by all distinct zeros of A_i except the marked zero. Their permutation under ϕ , together with (4.1), identifies the omitted root and proves (4.3). On the marked plane one has identically

$$p_i = -4, \quad Q_i = -16.$$

Specializing (4.7) gives $\rho^2 = \rho^3 = 1$, hence $\rho = 1$. Solving the first relation in (4.7) for $\hat{b}$, and the second for $\hat{a}$, shows that the coefficient of b in $\hat{a}$ is

$$\frac{B_1 - \tilde{B}_2}{6\kappa^2 A_1}.$$

Its polynomiality gives $A_1 \mid B_1 - \tilde{B}_2$, which is (4.2), including the residue jets at repeated roots.

If A_1 is linear, (4.4) has no plane component; the same is true of A_2 . An affine ϕ matches the marked roots and gives (4.1); (4.2) follows from $B_i(\alpha_i) = -2$. $\square$

4.3. Sufficiency and an explicit lift.

Proof of sufficiency in theorem 4.1. Assume (4.1)–(4.3). Put

$$\lambda = \kappa^{-1}, \quad q(c) = \frac{\lambda(B_1(c) - B_2(\phi(c)))}{3A_1(c)} \in \mathbb{C}[c].$$

Define the triangular target automorphism

$$c_2 = \phi(c), \tag{4.8}$$

$$b_2 = \lambda b + \frac{3A_1 q^2}{\lambda} - 2B_1 q, \tag{4.9}$$

$$a_2 = \lambda^2 a + \frac{\lambda q}{2} b + \frac{A_1 q^3}{2\lambda} - \frac{B_1 q^2}{2}. \tag{4.10}$$

Direct coefficient comparison gives

$$P_2(\lambda T + q; a_2, b_2, \phi(c)) = \lambda^2 P_1(T; a, b, c). \tag{4.11}$$

Let $w_i = w_{A_i, B_i, \alpha_i}$ and define

$$X = \frac{x}{\lambda}, \quad Y = \lambda y + q(c), \quad Z = \frac{\lambda^3}{x^3} (w_2(X, Y) - \phi(c)). \tag{4.12}$$

This is polynomial. Indeed, if $q_0 = q(\alpha_1)$, differentiating (4.1) and

$$B_2(\phi(c)) = B_1(c) - \frac{3A_1(c)q(c)}{\lambda}$$

at $c = \alpha_1$ gives the two-jet identity

$$w_2\left(\frac{x}{\lambda}, \lambda y + q_0\right) = \phi(w_1(x, y)).$$

Moreover $c - \alpha_1 \in (x)$, so $q(c) - q_0 \in (x)$; the variable Y occurs in w_2 multiplied by X^2 , and replacing q_0 by $q(c)$ therefore changes the left side by a multiple of x^3 . Finally $w_1 - c = x^3z$, proving divisibility of the numerator in (4.12).

The same construction with

$$\phi^{-1}, \quad \lambda^{-1}, \quad q_{\text{inv}}(c_2) = -\lambda^{-1}q(\phi^{-1}(c_2))$$

is a polynomial inverse. Thus (4.12) is a source automorphism. Since

$$Y + \frac{1}{X} = \lambda\left(y + \frac{1}{x}\right) + q(c),$$

(4.11) and its derivative give

$$F_2(X, Y, Z) = \Psi(F_1(x, y, z))$$

on $x \neq 0$, hence everywhere. $\square$

Corollary 4.4. *The equivalence classes are the $\text{Aff}_1(\mathbb{C})$ -orbits of triples*

$$(D, \alpha, \overline{B}),$$

where $D = \text{div}(A)$ is an effective divisor on $\mathbb{A}^1$, α is a marked multiplicity-one point of D , and

$$\overline{B} \in \mathbb{C}[c]/(A), \quad \overline{B}(\alpha) = -2.$$

For degree d in a multiplicity stratum with $r \geq 2$ distinct roots, the generic dimension is $d + r - 3$; in the squarefree stratum it is $2d - 3$.

5. STABLE UNIQUENESS IN THE TWO-BLOCK INCIDENCE FAMILY

Let

$$V_d = H^0(\mathbb{P}^1, \mathcal{O}(d))$$

and consider multiplication of binary forms

$$\mu_{a,b}: \mathbb{P}(V_a) \times \mathbb{P}(V_b) \longrightarrow \mathbb{P}(V_{a+b}), \quad ([f], [g]) \longmapsto [fg].$$

Let $R_{a,b}$ be the resultant divisor. Fix the point $[Y^{a+b}]$ on the rational normal curve, and let H be any hyperplane tangent but not osculating there. In coefficient coordinates

$$h = \sum_{i=0}^{a+b} h_i X^i Y^{a+b-i},$$

its functional can be normalized to

$$\lambda(h) = h_2 + \lambda_3 h_3 + \cdots + \lambda_{a+b} h_{a+b}. \quad (5.1)$$

Define

$$U_{a,b,H} = (\mathbb{P}(V_a) \times \mathbb{P}(V_b)) \setminus (R_{a,b} \cup \mu_{a,b}^{-1}(H)). \quad (5.2)$$

Theorem 5.1 (Stable uniqueness). *Let $a, b \geq 1$ and $a + b \geq 3$. For every hyperplane H tangent but not osculating at the chosen point,*

$$U_{a,b,H} \simeq \mathbb{A}^{a+b} \iff \{a, b\} = \{1, 2\}.$$

More strongly, outside the cubic case,

$$U_{a,b,H} \times \mathbb{A}^r \not\simeq \mathbb{A}^{a+b+r} \quad \text{for every } r \geq 0.$$

5.1. The class-group reduction. Put $D_H = \mu_{a,b}^{-1}(H)$. The matrix of the bilinear form $(f, g) \mapsto \lambda(fg)$ has rank at least two, so D_H is irreducible. The two removed prime divisors have classes

$$[R_{a,b}] = (b, a), \quad [D_H] = (1, 1)$$

in

$$\text{Cl}(\mathbb{P}^a \times \mathbb{P}^b) = \mathbb{Z}^2.$$

Divisor localization gives

$$\text{Cl}(U_{a,b,H}) \simeq \mathbb{Z}^2 / \langle (b, a), (1, 1) \rangle. \quad (5.3)$$

Thus a trivial class group forces $|a - b| = 1$. Since the class group of these smooth varieties is unchanged by adjoining polynomial variables, (5.3) also obstructs stable affine space unless the degrees are adjacent.

It remains to take $b = a + 1$, $a \geq 2$, and put

$$n = 2a + 1, \quad X = \mathbb{P}^a \times \mathbb{P}^{a+1}, \quad C = X \setminus R_{a,a+1}.$$

Here we have interchanged the two factors if necessary.

5.2. The coprime-pair locus. Write

$$P_m = [\mathbb{P}^m] = 1 + \mathbb{L} + \cdots + \mathbb{L}^m.$$

Stratifying pairs of effective divisors on $\mathbb{P}^1$ by the degree of their gcd gives motivic Möbius inversion. Since

$$\sum_{d \geq 0} P_d T^d = \frac{1}{(1 - T)(1 - \mathbb{L}T)},$$

we obtain

$$\begin{aligned} [C] &= P_a P_{a+1} - (1 + \mathbb{L}) P_{a-1} P_a + \mathbb{L} P_{a-2} P_{a-1} \\ &= \mathbb{L}^n + \mathbb{L}^{n-1}. \end{aligned} \quad (5.4)$$

5.3. The hyperplane divisor. Let r be the rank of the matrix of $(f, g) \mapsto \lambda(fg)$. It contains the minor

$$\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & \lambda_3 \\ 1 & \lambda_3 & \lambda_4 \end{pmatrix}, \quad \det = -1,$$

so $r \geq 3$. Projecting D_H to $\mathbb{P}^a$, its fiber is $\mathbb{P}^a$ away from the left radical and $\mathbb{P}^{a+1}$ on the radical $\mathbb{P}^{a-r}$. Therefore

$$[D_H] = P_a^2 + \mathbb{L}^{a+1} P_{a-r} \equiv \mathbb{L}^{n-1} + 2\mathbb{L}^{n-2} \pmod{F_{n-3}K_0}. \quad (5.5)$$

We use the convention $P_m = 0$ for $m < 0$.

5.4. The codimension-two intersection. Resolve the resultant by marking a common linear factor:

$$\tilde{R} = \mathbb{P}^1 \times \mathbb{P}^{a-1} \times \mathbb{P}^a \longrightarrow R_{a,a+1}, \quad (\ell, [f_1], [g_1]) \longmapsto ([\ell f_1], [\ell g_1]).$$

The inverse image of D_H is

$$\tilde{Z} = \left\{ \lambda(\ell^2 f_1 g_1) = 0 \right\}. \quad (5.6)$$

At $\ell = Y$, the residual bilinear matrix contains

$$\begin{pmatrix} 0 & 1 \\ 1 & \lambda_3 \end{pmatrix},$$

so its generic rank is at least two and its generic defining polynomial is irreducible.

There are no vertical components. Write $\ell = vY + uX$. The coefficients of the restricted functional are

$$\rho_j = v^2 \lambda_j + 2uv \lambda_{j+1} + u^2 \lambda_{j+2}.$$

If all ρ_j vanished, then $\rho_0 = u^2 \lambda_2 = 0$ would give $u = 0$, while $\rho_2 = v^2 \lambda_2 \neq 0$, a contradiction. Gauss's lemma therefore shows that $\tilde{Z}$ is geometrically irreducible.

On a generic affine chart, its bilinear equation can be solved for one residual coefficient, so its function field is rational. The locus where f_1 and g_1 have an additional common factor is the residual resultant divisor. On the generic $\mathbb{P}^{a-1} \times \mathbb{P}^a$ fiber it has class $(a, a-1)$, whereas $\tilde{Z}$ has class $(1, 1)$. The two irreducible divisors therefore differ, so the additional-common-factor locus does not contain $\tilde{Z}$. Hence the map to

$$Z = R_{a,a+1} \cap D_H$$

is generically one-to-one. Consequently Z is irreducible and rational of dimension $n-2$, and

$$[Z] \equiv \mathbb{L}^{n-2} \pmod{F_{n-3}K_0}. \quad (5.7)$$

5.5. Proof of stable uniqueness.

Proof of theorem 5.1. For the adjacent higher pairs, inclusion–exclusion and (5.4), (5.5), and (5.7) give

$$[U_{a,a+1,H}] \equiv \mathbb{L}^n - \mathbb{L}^{n-2} \pmod{F_{n-3}K_0}. \quad (5.8)$$

Applying the compactly supported Hodge–Deligne polynomial, terms in $F_{n-3}K_0$ cannot change the coefficient of $(uv)^{n-2}$. Hence

$$E_c(U_{a,a+1,H}; u, v) \neq (uv)^n = E_c(\mathbb{A}^n; u, v).$$

Multiplication by $\mathbb{A}^r$ multiplies the polynomial by $(uv)^r$, so the same deficit rules out every stable affine-space isomorphism.

For nonadjacent pairs the class-group obstruction already applies. In the cubic case $\{a, b\} = \{1, 2\}$, the public marked-root construction gives $U_{a,b,H} \simeq \mathbb{A}^3$; the stabilizer of the chosen point on the rational normal cubic is transitive on the tangent nonosculating hyperplanes. $\square$

Remark 5.2 (Why the cubic is the knife edge). For $a = 1$, the residual product in (5.6) has degree one. At the tangent point, the restricted functional vanishes on an entire residual fiber, giving a second top-dimensional component of $R \cap D_H$. Its extra $\mathbb{L}^{n-2}$ contribution cancels the deficit in (5.8). For $a \geq 2$, the coefficient $\rho_2 = v^2 \lambda_2$ prevents that vertical component. The geometric transition is therefore a codimension-two component count, not merely the special solvability of cubic equations.

6. THE CONDITIONAL UNIVERSAL COVER AND THE REMAINING DEFECT

These results do not show that every generic-degree-three Keller map belongs to the normalized family. There is, however, a useful conditional universal statement.

Let $V = \text{Sym}^3(\mathbb{C}^2)^\vee$. The space of a linear form L , a quadratic form Q , and the normalization $\text{Res}(L, Q) = 1$ is isomorphic to

$$\text{SL}_2 \times \mathbb{A}^1.$$

Multiplication $(L, Q) \mapsto LQ$ is the universal marked-simple-root map over the simple-root locus. This description is compatible with the classical binary-form and flat-triple-cover frameworks [Mir85, Woo11].

Proposition 6.1 (Conditional master-cover statement). *Let $F: \mathbb{A}^3 \rightarrow \mathbb{A}^3$ be a generic-degree-three Keller map, and let $\overline{X} \rightarrow \mathbb{A}^3$ be the normalization of the target in its function field. If $\overline{X} \rightarrow \mathbb{A}^3$ is finite flat of degree three, then over the simple-root locus it is a pullback of the universal resultant-one marked-root cover. Recovering the original affine source from $\overline{X}$ additionally requires a theorem identifying the entire deleted boundary.*

Proof sketch and scope. After base change to the source function field, the chosen sheet splits off from the cubic algebra and leaves a quadratic algebra. Finite flatness makes the trace-zero part locally free of rank two, so the multiplication data are encoded by a binary cubic, equivalently by the classical triple-cover data. Resultant normalization then gives the map to the universal factor space. This identifies the finite cover on the simple-root locus. It does not prove that the given affine source deletes exactly the ramification and expected infinity components; unramified sheets may also be absent. That boundary-completeness statement is a separate hypothesis. $\square$

Question 6.2 (Cubic flatness). *Does the Keller condition force the finite cubic normalization $\overline{X} \rightarrow \mathbb{A}^3$ to be flat?*

The trace-zero summand is a rank-two reflexive module, so a possible defect is confined to finitely many target points. General normal S_3 -covers

need not be flat. We know no argument that rules out this defect using only the Keller condition. Thus question 6.2, rather than the already understood universal marked-root algebra, is the current obstruction to a general cubic classification.

Question 6.3 (Boundary Torelli). *To what extent is a marked-root cover determined, up to polynomial left–right equivalence, by its deleted boundary together with the induced normalization, conductor, and incidence data? In particular, which hypotheses make the boundary-completeness problem in proposition 6.1 a reconstruction theorem?*

A positive answer would turn the nonproperness boundary from a defect of the affine map into usable classification data. The normalized A, B family and the stable-uniqueness calculation provide test cases, but no general boundary Torelli theorem is claimed here.

7. EXACT COMPUTATIONS AND REPRODUCIBILITY

Three exact Python/SymPy programs accompany the internal evidence record. They use rational or finite-field arithmetic only.

- (1) The normalized-family verifier checks the forced jets in theorem 3.1, the restriction to $x = 0$, the determinant, the universal discriminant, (3.11), and a multi-root example with explicit collisions and two deleted infinity planes.
- (2) The equivalence stress verifier checks (4.11), the cusp identity (4.6), and the complete source–target conjugacy for a repeated-root example. It does not prove the Blanc–Stampfli input or the necessity argument.
- (3) The stable-uniqueness verifier checks the motivic polynomial identities and exact point counts for all tangent hyperplanes in the tested $(2, 3)$ and $(3, 4)$ finite-field cases. It does not prove the class-group, irreducibility, rationality, or Hodge–Deligne steps.

The reviewed source files are included in `code/core/`; the root-slope checks are in `code/verify_root_slope_extensions.py`. The accompanying `COMPUTATION.md` gives a claim-to-file index and exact run commands. Fresh logs and a hash manifest are distributed in the versioned computational supplement. These calculations expose the algebra behind the normal forms: pole cancellation fixes the first two transverse jets, the cubic discriminant factors on the chosen-root chart, and the same covariants control source–target equivalence. They do not replace the geometric arguments about normalization, boundary components, or automorphism groups.

APPENDIX A. ROOT–SLOPE GEOMETRY AND MONODROMY

This appendix isolates the mechanism behind the coordinate frame. It also records two consequences that complete the generic covering picture. The calculations are independent of the pole-cancellation classification.

Proposition A.1 (Universal root-slope transform). *Let $H(T, c) \in k[T, c]$, where $2 \in k^*$, and define*

$$b = r - H_T(t, c), \quad 2a = H(t, c) + tb.$$

Then, for

$$P_{a,b,c}(T) = H(T, c) + bT - 2a,$$

one has

$$P_{a,b,c}(t) = 0, \quad P'_{a,b,c}(t) = r,$$

and

$$2 da - t db - H_c(t, c) dc = r dt.$$

Consequently

$$\det \frac{\partial(a, b, c)}{\partial(t, r, c)} = \frac{r}{2}.$$

Proof. The root and derivative identities are immediate from the definitions. Differentiating $2a = H(t, c) + tb$ and substituting $H_T(t, c) + b = r$ gives the one-form identity. Wedge it with $db \wedge dc$; since

$$db = dr - H_{TT} dt - H_{Tc} dc,$$

the stated determinant follows. $\square$

For the reciprocal chart

$$t = y + \frac{1}{x}, \quad r = \frac{2}{x}, \quad c = w(x, y) - x^3 z,$$

direct expansion gives

$$\det \frac{\partial(t, r, c)}{\partial(x, y, z)} = -2x = -\frac{4}{r}.$$

Thus the composite has determinant -2 wherever it is defined. If pole cancellation makes it polynomial across $x = 0$, it is a Keller map on all of affine space.

Remark A.2 (The mechanism). The finite root-forgetting map contributes the vanishing factor $r = P'(t)$. The affine chart contributes its reciprocal. When two roots collide, $P'(t)$ tends to zero and $x = 2/P'(t)$ tends to infinity. The ramification point is not canceled inside the affine source; it is deleted from the affine chart. This is the precise gap between an everywhere étale affine map and a finite étale cover.

Proposition A.3 (Marked Vandermonde factorization). *If $P(T) = (T - t)Q(T)$, then*

$$\text{Disc}(P) = P'(t)^2 \text{Disc}(Q).$$

For

$$P(T) = A(c)T^3 + B(c)T^2 + bT - 2a, \quad r = P'(t),$$

this becomes

$$\text{Disc}(P) = r^2((3A(c)t + B(c))^2 - 4A(c)r).$$

Proof. Use the product formula

$$\text{Disc}((T - t)Q) = \text{Disc}(T - t) \text{Disc}(Q) \text{Res}(T - t, Q)^2$$

and $\text{Res}(T - t, Q) = Q(t) = P'(t)$. Dividing the cubic by $T - t$ gives a quadratic whose discriminant is the displayed residual factor. $\square$

The two factors represent different collisions. The first records a collision involving the marked root; the second records collision of the two unmarked roots. Only the first ramifies the chosen sheet.

Theorem A.4 (Generic S_3 monodromy). *Assume characteristic zero and generic degree three. The normal closure of the normalized cubic extension has Galois group S_3 . Over $\mathbb{C}$, the geometric monodromy of the three-sheeted étale locus is therefore S_3 .*

Proof. Generic degree three makes the inverse cubic irreducible over $\bar{k}(a, b, c)$. Its discriminant is

$$\Delta = B^2b^2 - 4Ab^3 + 8aB^3 - 36aABb - 108a^2A^2.$$

As a quadratic in a ,

$$\text{Disc}_a(\Delta) = -64(3Ab - B^2)^3.$$

The right side is not a square in $\bar{k}(b, c)$, so Δ is not a square in the rational function field. An irreducible cubic with nonsquare discriminant has Galois group S_3 . $\square$

On $A(c) \neq 0$, depressing the cubic gives coordinates

$$p = \frac{3Ab - B^2}{3A^2}, \quad q = \frac{2B^3 - 9ABb - 54A^2a}{27A^3}$$

for which

$$\Delta = A(c)^4(-4p^3 - 27q^2).$$

Hence

$$\{A(c) \neq 0, \Delta \neq 0\} \cong (\mathbb{A}_c^1 \setminus V(A)) \times \{(p, q) : 4p^3 + 27q^2 \neq 0\}.$$

If A has s distinct complex roots, the fundamental group is

$$F_s \times B_3.$$

The F_s -factor records deleted infinity levels; the braid factor gives the standard permutation quotient $B_3 \twoheadrightarrow S_3$.

Verification boundary. The accompanying verifier checks the root-slope Jacobian, reciprocal chart, discriminant identities, and the finite motivic identities in the companion research note. The full companion source and its generated PDF are retained under **supplements/**; the present appendix includes only the results integrated into this manuscript.

APPENDIX B. OMITTED VALUES AND THE SINGULAR NONPROPERNESS
LOCUS

This appendix records a general consequence that arose from the cubic incidence calculation but does not depend on generic degree three. It is included as a theorem pending an independent check of the standard purity and tame-local-form inputs.

Let

$$F: X = \mathbb{A}_{\mathbb{C}}^n \longrightarrow Y = \mathbb{A}_{\mathbb{C}}^n$$

be a Keller map. Write

$$O_F = Y \setminus F(X)$$

and let S_F be the nonproperness set, with its reduced structure.

Theorem B.1 (Omitted values are singular). *For every complex polynomial Keller map,*

$$O_F \subseteq \text{Sing}(S_F).$$

Proof. Let

$$\pi: \tilde{X} \longrightarrow Y$$

be the normalization of Y in $\mathbb{C}(X)$. The map factors as an open immersion $j: X \hookrightarrow \tilde{X}$ followed by the finite map π . Put $D = \tilde{X} \setminus X$. The complement of the affine open X is pure of codimension one, and

$$S_F = \pi(D).$$

First, O_F has codimension at least two. Indeed, if an irreducible divisor $V(h) \subset Y$ were omitted, then $h \circ F$ would be a nowhere-zero polynomial on affine space and hence a nonzero constant. Dominance makes F^* injective, forcing h itself to be constant.

Now take $y \in S_F^{\text{reg}}$. After shrinking and making an étale base change, the finite normalization has the standard tame local form

$$\text{Spec } A[u_i]/(u_i^{e_i} - f) \longrightarrow \text{Spec } A,$$

where $S_F = V(f)$. The unique divisor over S_F in the i -th piece is $E_i = V(u_i)$. Purity of D prevents deleting only an isolated codimension-two subset of a retained sheet.

Every sheet with $e_i > 1$ must be deleted: retaining its divisor in X would give ramification of the Keller map. On the other hand, not all divisors E_i can be deleted, since then a dense open part of S_F would be omitted, contradicting $\text{codim } O_F \geq 2$. Thus an unramified sheet with $e_i = 1$ is retained, and its point over y belongs to X . Hence $y \in F(X)$, proving the claim. $\square$

Corollary B.2. *If S_F is smooth, then F is surjective.*

Proof. The singular locus is empty, so theorem B.1 gives $O_F = \emptyset$. $\square$

For the normalized cubic family of this paper, singularity is visible directly. At any c_0 with $A(c_0) \neq 0$, the cubic

$$P_{a,b,c_0}(T) = A(c_0)T^3 + B(c_0)T^2 + bT - 2a$$

has a unique triple-root specialization

$$t_0 = -\frac{B(c_0)}{3A(c_0)}, \quad b_0 = \frac{B(c_0)^2}{3A(c_0)}, \quad a_0 = -\frac{B(c_0)^3}{54A(c_0)^2}.$$

After translating $T = t_0 + W$, the local discriminant is the cusp $-4p^3 - 27q^2$. Consequently the reduced nonproperness hypersurface of every admissible member of the A, B family is singular along its triple-root curve.

Remark B.3 (Scope). The theorem concerns the reduced nonproperness set, not the critical-value locus: a Keller map has no critical points. No claim of priority is made until the general statement and its tame-local-form argument have received independent review.

APPENDIX C. DISCRIMINANT-DEPENDENT HYPERPLANES

This appendix treats the first nonlinear deformations of the hyperplane that cuts out the marked-root model. It is logically independent of the classification in the body of the paper and is retained as a self-contained working argument.

Write

$$L = aX + bY, \quad M = -qX + pY, \quad ap + bq = 1,$$

and

$$\Phi_{G,\gamma} = L(M^2 + \gamma L^2).$$

The normalized factor space is $SL_2 \times \mathbb{A}_\gamma^1$, and its coefficient map to binary cubics is

$$\begin{aligned} A_0 &= a^3\gamma + aq^2, \\ A_1 &= 3a^2b\gamma - 2apq + bq^2, \\ A_2 &= 3ab^2\gamma + ap^2 - 2bpq, \\ A_3 &= b^3\gamma + bp^2. \end{aligned}$$

Its discriminant is $\text{Disc}(\Phi_{G,\gamma}) = -4\gamma$.

Let $\lambda(t)$ be a polynomial family of nonzero covectors whose generic projective class is tangent but not osculating to the twisted cubic. For $f \in \mathbb{C}[t]$, define the full marked-simple-root incidence by

$$X_{\lambda,f} = \{(G, \gamma) : \lambda(\gamma)(\Phi_{G,\gamma}) = f(\gamma)\}.$$

Theorem C.1 (Moving-tangent rigidity). *If $X_{\lambda,f} \simeq \mathbb{A}^3$, then f is a nonzero constant, $\lambda(t)$ has tangent nonosculating type for every t , and there is an $h(t) \in SL_2(\mathbb{C}[t])$ carrying $\lambda(t)$ to a fixed tangent normal. The resulting incidence projection is polynomially left-right equivalent to the original marked-root map. Conversely, these conditions give that class.*

Proof. At a fixed value $t = \gamma$, put the normal into tangent form A_2 or osculating form A_3 . Stratification of the corresponding hypersurface in SL_2 gives the compactly supported Euler table

	$t \neq 0, c \neq 0$	$t \neq 0, c = 0$	$t = 0, c \neq 0$	$t = 0, c = 0$
tangent (2, 1)	3	0	1	0
osculating (3)	-3	0	0	0
zero normal	0	0	0	0

For example, in tangent form

$$A_2 = 3a(p^2 + b^2t) - 2p.$$

The stratum $b = 0$ contributes one when the level is nonzero. On $b \neq 0$, the divisor $p^2 + b^2t = 0$ consists, for $t \neq 0$, of two copies of $\mathbb{G}_m$; the two exceptional centers are replaced by affine lines, giving the remaining contribution two. In osculating form,

$$A_3 = b(p^2 + b^2t),$$

and the nonzero-level fiber is $\mathbb{A}^1$ times a genus-one double cover of $\mathbb{P}_b^1$ with three punctures, hence has Euler characteristic -3 .

Euler integration over $\mathbb{A}_t^1$ now gives $\chi_c(X_{\lambda,f}) \leq 1$. Equality is possible only when f has no zero, the normal never degenerates, and the fiber over $t = 0$ is tangent at nonzero level. Thus $f \in \mathbb{C}^*$ and every specialization has type (2, 1).

Represent the normal as a binary cubic $\Lambda(t; U, V)$. Since it has type (2, 1) everywhere, Gauss's lemma factors it as

$$\Lambda = rL^2M$$

with polynomial linear forms L, M , where r and $\det(L, M)$ are nonzero constants. Their coefficient rows therefore give $h(t) \in SL_2(\mathbb{C}[t])$. The target transformation

$$\Psi_h(\Phi) = h(-\text{Disc}(\Phi)/4) \cdot \Phi$$

is a polynomial automorphism because the discriminant is SL_2 -invariant. Equivariance supplies the matching source automorphism and reduces the equation to a fixed tangent hyperplane at nonzero constant level. $\square$

The infinitesimal SL_2 -orbit of the tangent normal A_2 spans A_1, A_2, A_3 ; the first transverse coefficient is A_0 . The first nonlinear transverse family is therefore

$$X_\kappa = \{A_2 + \kappa\gamma A_0 = c\} \subset SL_2 \times \mathbb{A}^1, \quad \kappa, c \in \mathbb{C}^*.$$

Theorem C.2 (First transverse obstruction). *For every $\kappa, c \neq 0$,*

$$\chi_c(X_\kappa) = -5.$$

In particular, $X_\kappa \not\cong \mathbb{A}^3$.

Proof. On $a = 0$, one has $bq = 1$, $A_0 = 0$, and $A_2 = -2p$; this stratum is $\mathbb{G}_m \times \mathbb{A}^1$ and contributes zero. On $a \neq 0$, set

$$s = b/a, \quad z = aq, \quad u = 3s^2 + \kappa t.$$

After multiplying by a , the equation is quadratic in z :

$$uz^2 - 4sz + 1 + a^4tu - ac = 0.$$

The locus $u = 0$ contributes one. On $u \neq 0$, the quadratic cover has base Euler characteristic zero, so its contribution is minus that of its branch surface. After putting $w = a^2u/\sqrt{\kappa}$, the branch equation is controlled inside $(\mathbb{G}_m)_{a,w}^2$ by

$$D_A = V(4 + 3w^2), \quad D_C = V(1 - ac + w^2).$$

Here

$$\chi_c(D_A) = 0, \quad \chi_c(D_C) = -2, \quad \chi_c(D_A \cap D_C) = 2.$$

The two-sheet, one-sheet, empty, and affine-line fibers over the four resulting strata give branch-surface Euler characteristic 6. Consequently $\chi_c(X_\kappa) = 1 - 6 = -5$. $\square$

For a general polynomial $r(t)$, the same calculation reduces

$$\{A_2 + r(\gamma)A_0 = c\}$$

to an explicit branch-surface Euler problem. If $n(r)$ is the number of distinct roots of r , a necessary condition for affine-three-space source is

$$\chi_c(\mathcal{B}_r) = n(r) - 1.$$

Determining whether any higher r satisfies this condition remains open.

APPENDIX D. COMMON ZEROS, MAXIMAL CUBIC ORDERS, AND THE MISSING LOCAL MODEL

The squarefree and coprime locus already suffices for the principal classification in the body. When A and B have a common zero, however, the primitive root coordinate need not give the finite normalization. This appendix gives the correction and an explicit example with two unramified sheets deleted along one divisor.

D.1. The corrected cubic order. Let

$$R = \mathbb{C}[a, b, c], \quad P(T) = A(c)T^3 + B(c)T^2 + bT - 2a,$$

and let t denote a root of P in the generic cubic extension. Put

$$\omega = A(c)t, \quad \theta = A(c)t^2 + B(c)t.$$

Then

$$\omega^2 = A\theta - B\omega, \quad \omega\theta = 2Aa - b\omega, \quad \theta^2 = 2aB + 2a\omega - b\theta. \quad (\text{D.1})$$

Thus $R \oplus R\omega \oplus R\theta$ is the primitive finite cubic order.

For every zero β of A , write

$$m_\beta = \text{ord}_\beta A, \quad n_\beta = \text{ord}_\beta B$$

and set

$$k_\beta = \min \{n_\beta, \lfloor m_\beta/2 \rfloor\}, \quad H(c) = \prod_{\beta \in Z(A)} (c - \beta)^{k_\beta}.$$

Equivalently, H is the largest monic polynomial such that

$$H^2 \mid A, \quad H \mid B.$$

Write

$$\bar{A} = A/H^2, \quad \bar{B} = B/H, \quad q = \omega/H.$$

Theorem D.1 (Maximal-order correction). *The integral closure of R in the generic cubic extension is*

$$\mathcal{C}^{\text{nor}} = R \oplus Rq \oplus R\theta,$$

with multiplication

$$q^2 = \bar{A}\theta - \bar{B}q, \tag{D.2}$$

$$q\theta = 2aH\bar{A} - bq, \tag{D.3}$$

$$\theta^2 = 2aH\bar{B} + 2aHq - b\theta. \tag{D.4}$$

It is finite flat of degree three over R . If $\mathcal{D}$ is the primitive binary-cubic discriminant, then

$$\text{Disc}(\mathcal{C}^{\text{nor}}) = \mathcal{D}/H^2.$$

Proof. The identities (D.2)–(D.4) follow from (D.1); hence the displayed module is a finite free cubic order. The change from the basis $(1, q, \theta)$ to $(1, \omega, \theta)$ has determinant H , giving the discriminant formula.

It remains to prove maximality. At a zero β of A , the $(c - \beta)$ -content of

$$\mathcal{D} = B^2b^2 - 4Ab^3 + 8aB^3 - 36aABb - 108a^2A^2$$

has order

$$\delta_\beta = \min(m_\beta, 2n_\beta).$$

After division by H^2 , the residual exponent is

$$\delta_\beta - 2k_\beta = \begin{cases} 1, & m_\beta < 2n_\beta \text{ and } m_\beta \text{ is odd,} \\ 0, & \text{otherwise.} \end{cases}$$

The nonvertical factor is also reduced: as a quadratic in a , its discriminant is

$$-64(3Ab - B^2)^3,$$

which is not a square in $\mathbb{C}(c, b)$. Thus $\mathcal{D}/H^2$ is squarefree.

For a cubic order over a discrete valuation ring, the discriminant of a proper suborder differs from that of the maximal order by the square of its index. The squarefree discriminant therefore makes every height-one localization of $\mathcal{C}^{\text{nor}}$ maximal. The algebra is free over the regular ring R ,

hence satisfies S_2 ; height-one maximality and Serre's criterion show that it is normal. It is consequently the integral closure. $\square$

D.2. The four generic divisor behaviors.

Proposition D.2 (Four generic divisor behaviors). *Let $\beta \neq \alpha$ be a zero of A , and retain the notation (m_β, n_β) . The generic behavior over the plane $c = \beta$ is as follows:*

<i>Condition</i>	<i>Completed fiber</i>	<i>Affine size</i>	<i>Inertia</i>
$B(\beta) \neq 0$	one simple infinity root deleted	2	1
$B(\beta) = 0$, $m_\beta < 2n_\beta$, with m_β odd	one ramified pair deleted	1	(12)
$B(\beta) = 0$, otherwise	two unramified sheets deleted	1	1

Together with the excluded generic three-cycle behavior, these are the four generic divisor types stated in remark 3.6.

Proof. In the reciprocal coordinate $u = 1/t$, the two roots approaching infinity satisfy

$$A_0 s^{m_\beta} + B_0 s^{n_\beta} u + bu^2 - 2au^3 = 0, \quad s = c - \beta.$$

If $m_\beta < 2n_\beta$, both valuations are $m_\beta/2$, and the pair ramifies exactly when m_β is odd. If $m_\beta = 2n_\beta$, the residual quadratic after $u = s^{n_\beta} v$ is

$$bv^2 + B_0 v + A_0.$$

If $m_\beta > 2n_\beta$, the two valuations are n_β and $m_\beta - n_\beta$. This proves the table and, in particular, separates the one-sheet and two-sheet unramified cases. $\square$

D.3. An explicit two-sheet-loss example.

Proposition D.3 (A U_2 Keller map). *Take*

$$A(c) = c(c-1)^2, \quad B(c) = 2(c-1), \quad \alpha = 0,$$

and

$$c = 2x + (6 - 3y)x^2 - x^3 z, \quad t = y + \frac{1}{x}.$$

Define

$$b = \frac{2}{x} - 3c(c-1)^2 t^2 - 4(c-1)t$$

and

$$2a = \frac{2t}{x} - 2c(c-1)^2 t^3 - 2(c-1)t^2.$$

The apparent poles cancel. The map $F = (a, b, c)$ is polynomial of component degrees $(15, 14, 4)$, has

$$\det DF = -2,$$

and has the collision

$$F(0, 4, -22) = F(1, -1, 11) = F(-1, 2, 2) = (0, 2, 0).$$

Along $c = 1$, the finite normalization has one retained sheet and two generically unramified deleted sheets.

Proof. Here $H = c - 1$, $\overline{A} = c$, and $\overline{B} = 2$. The normalized binary cubic is

$$cT^3 + 2ST^2 + bS^2T - 2a(c - 1)S^3.$$

At $c = 1$ it becomes

$$T(T^2 + 2ST + bS^2).$$

The root $T = 0$ is the retained affine sheet, while the quadratic factor gives two distinct sheets for generic b . Direct expansion gives the polynomiality, degrees, determinant, and displayed collision. $\square$

For this example the normalized discriminant is

$$\mathcal{D}_{\text{nor}} = -4(27a^2c^4 - 54a^2c^3 + 27a^2c^2 + 18abc^2 - 18abc - 16ac + 16a + b^3c - b^2),$$

and

$$\mathcal{D}_{\text{nor}}|_{c=1} = -4b^2(b - 1).$$

Consequently

$$S_F = V(\mathcal{D}_{\text{nor}}) \cup V(c - 1).$$

The line $V(c - 1, b)$ is omitted, whereas $V(c - 1, b - 1)$ is a branch/plane intersection that is not omitted. Hence

$$\mathbb{A}^3 \setminus F(\mathbb{A}^3) \subsetneq \text{Sing}(S_F).$$

This shows that the equality between omission and the singular locus seen in the base map is not a general feature of the cubic frame.

Remark D.4. The all-multiplicity ordinary left-right classification is already theorem 4.1. Stable recovery in the presence of common roots requires the weighted relative-Jacobian package developed in the stable moduli companion paper [Mon].

APPENDIX E. QUADRATIC COVARIANT RIGIDITY

We retain the universal marked-root coordinates

$$G = \begin{pmatrix} a & b \\ -q & p \end{pmatrix} \in \text{SL}_2, \quad L = aX + bY, \quad M = -qX + pY,$$

and

$$\Phi_{G,t} = L(M^2 + tL^2) = A_0X^3 + A_1X^2Y + A_2XY^2 + A_3Y^3.$$

Thus

$$\begin{aligned} A_0 &= a^3t + aq^2, \\ A_1 &= 3a^2bt - 2apq + bq^2, \\ A_2 &= 3ab^2t + ap^2 - 2bpq, \\ A_3 &= b^3t + bp^2, \end{aligned}$$

and $\text{Disc}(\Phi_{G,t}) = -4t$. The base construction is the full incidence over $A_2 = c$, with $c \neq 0$.

E.1. The Hessian summand. Write, up to a common nonzero scalar,

$$\text{Hess}(\Phi) = H_0 X^2 + H_1 XY + H_2 Y^2,$$

where

$$H_0 = 3A_0A_2 - A_1^2, \quad H_1 = 9A_0A_3 - A_1A_2, \quad H_2 = 3A_1A_3 - A_2^2.$$

For $\eta = (\eta_0, \eta_1, \eta_2)$, put

$$Q_\eta = \eta_0 H_0 + \eta_1 H_1 + \eta_2 H_2.$$

Theorem E.1 (Hessian exclusion). *For $\eta \neq 0$ and $\kappa \neq 0$, the target*

$$Y_{\eta, \kappa, c} = V(A_2 + \kappa Q_\eta - c)$$

and its full marked-root incidence preimage cannot both be isomorphic to $\mathbb{A}^3$.

Proof. As a quadratic form on the four-dimensional coefficient space,

$$\det Q_\eta = \frac{81}{16}(\eta_0\eta_2 - \eta_1^2)^2.$$

If this determinant is nonzero, translation removes the linear term A_2 , leaving a smooth affine quadric or a singular quadric cone, never $\mathbb{A}^3$. If $\eta_0\eta_2 - \eta_1^2 = 0$, write $\eta = (u^2, uv, v^2)$. The kernel is generated by

$$(-v^3, 3uv^2, -3u^2v, u^3).$$

When $uv = 0$, the target is a cylinder over an affine quadric surface. It remains only to consider $uv \neq 0$, for which the stabilizer of A_2 reduces to $\eta = (1, 1, 1)$.

Set

$$\ell = a + b, \quad r = b, \quad m = p - q, \quad s = p, \quad \ell s - rm = 1.$$

The incidence equation is linear in t , with coefficient

$$3D, \quad D = r^2(\ell - r) + \kappa\ell^2.$$

On $D \neq 0$, t is unique. On $D = 0$, put $x = r/\ell$; then

$$\ell = -\frac{\kappa}{x^2(1-x)}, \quad x \in \mathbb{A}^1 \setminus \{0, 1\}.$$

The compatibility equation is quadratic in m , with discriminant

$$-4\kappa^2(3x^4 - 4x^3 + 4\kappa c).$$

If ν is the number of roots of the quartic away from 0, 1, Euler integration gives

$$\chi_c(X_{\eta, \kappa, c}) = -1 - \nu.$$

It is therefore never $1 = \chi_c(\mathbb{A}^3)$. □

E.2. Rank-one squares. The quadratic representation decomposes as

$$\mathrm{Sym}^2(\mathrm{Sym}^3 \mathbb{C}^2) \simeq \mathrm{Sym}^6 \mathbb{C}^2 \oplus \mathrm{Sym}^2 \mathbb{C}^2.$$

The second summand is the Hessian component. The rank-one locus in the Cartan component consists of squares $E_\xi(\Phi)^2$, where $E_\xi(\Phi) = \Phi(\xi)$.

Theorem E.2 (Rank-one square classification). *For*

$$A_2 + \kappa E_\xi(\Phi)^2 = c,$$

one endpoint orbit is polynomially gauge-equivalent to the base construction from a tangent hyperplane. The other endpoint has incidence Euler characteristic -1 or 0 . The open orbit has Euler characteristic -1 or -2 . Hence no inequivalent full-incidence $\mathbb{A}^3$ occurs.

Proof. For the endpoint $E_\xi = A_3$, the target shear

$$\Phi(X, Y) \mapsto \Phi\left(X, Y + \frac{\kappa A_3}{3} X\right)$$

sends A_2 to $A_2 + \kappa A_3^2$ and lifts to the factor frame. For the endpoint $E_\xi = A_0$, the incidence equation is quadratic in t . Its branch equation is

$$4\kappa c \ell^4 - 16\kappa \ell^3 m r - 4\kappa \ell^3 + 9r^4 = 0,$$

which gives $\chi_c = -1$ for $c \neq 0$ and 0 for $c = 0$. For the open orbit, adapted coordinates reduce the branch equation to one generic stratum and the two exceptional values $x = 0, 2/3$, giving

$$\chi_c = -\mathbf{1}_{\{c \neq 0\}} - \mathbf{1}_{\{81\kappa c + 4 \neq 0\}}.$$

□

E.3. Products of distinct evaluations.

Theorem E.3 (Two-evaluation exclusion). *For distinct $\xi, \eta \in \mathbb{P}^1$, every $\kappa \neq 0$, and every c ,*

$$Y_{\xi, \eta} = V(A_2 + \kappa E_\xi E_\eta - c) \simeq \mathbb{A}^3,$$

but its full marked-root preimage $X_{\xi, \eta}$ is never $\mathbb{A}^3$.

Proof. The linear forms A_2, E_ξ, E_η are independent. Indeed, a binary cubic tangent to, but not osculating, the rational normal curve cannot be a sum of two cubes. Thus the target equation solves for A_2 .

The stabilizer of A_2 has four types of unordered distinct pairs:

$$\{0, \infty\}, \quad \{0, 1\}, \quad \{\infty, 1\}, \quad \{1, \lambda\}, \quad \lambda \notin \{0, 1\}.$$

Exact Euler integration in adapted determinant-one frames gives

$$\begin{aligned} \chi_c(X_{0, \infty}) &= 0, \\ \chi_c(X_{0, 1}) &= \begin{cases} -2, & \kappa c = 1/2, \\ -3, & \kappa c = 0 \text{ or } 64/3, \\ -4, & \text{otherwise,} \end{cases} \end{aligned}$$

and

$$\chi_c(X_{\infty,1}) = \begin{cases} 0, & \kappa c = -1/3, \\ -1, & \text{otherwise.} \end{cases}$$

For the open pair, the final branch curve has the form

$$lP_{\lambda,\kappa c}(x) = \kappa x Q_{\lambda}(x),$$

with both P and Q nonzero, and

$$\chi_c(X_{1,\lambda}) = -\#(Z(PQ) \cap \mathbb{C}^*) \leq 0.$$

None of these values is 1. $\square$

Remark E.4 (Representation-theoretic correction). The raw product $E_{\xi}E_{\eta}$ generally has components in both irreducible summands. For example,

$$10A_0A_3 = (A_0A_3 + A_1A_2) + (9A_0A_3 - A_1A_2).$$

Thus theorem E.3 is a natural quadratic-layer classification, not a classification of the pure Cartan secant variety.

E.4. The pure Cartan boundary.

Proposition E.5 (Catalecticant target obstruction). *The pure Cartan element*

$$C_{3,3} = A_0A_3 + A_1A_2$$

gives a nondegenerate affine quadric, not $\mathbb{A}^3$. More generally, a pure Cartan quadratic whose middle catalecticant is nonsingular is excluded at the target level.

Proof. After replacing A_1 by $A_1 + 1/\kappa$, the equation

$$A_2 + \kappa C_{3,3} = c$$

becomes

$$\kappa(A_0A_3 + A_1A_2) = c.$$

For $c \neq 0$ this is a smooth affine quadric threefold, of compactly supported Euler characteristic zero; for $c = 0$ it is singular. The same argument applies whenever the associated quadratic form, equivalently the middle catalecticant, is nonsingular. $\square$

Proposition E.6 (First rank-two Cartan endpoint). *For $\rho\kappa \neq 0$, the full incidence over*

$$A_2 + \kappa(A_0^2 + \rho A_3^2) = c$$

is not $\mathbb{A}^3$.

Proof. On the chart $a \neq 0$, put

$$s = b/a, \quad z = aq, \quad U = z^2 + a^4t.$$

The equation is quadratic in a :

$$-ca^2 + B(s, z, U)a + \kappa R(s, z, U) = 0,$$

where

$$B = 1 - 4sz + 3s^2U, \quad R = U^2 + \rho s^2(1 - 2sz + s^2U)^2.$$

For $c \neq 0$, the discriminant fibers degenerate at the nonempty zero set of

$$\rho s^6 + 4c\kappa \rho s^2 + 4.$$

If n_H is the number of its distinct roots, inclusion–exclusion gives

$$\chi_c(X) = -1 - n_H \leq -2.$$

For $c = 0$, the equation is linear in a and the corresponding calculation gives $\chi_c(X) = -6$. $\square$

Question E.7 (Remaining Cartan strata). *Classify the remaining rank-two Cartan secant orbits and the rank-three catalecticant stratum. The calculations above settle the full moving-linear layer and the principal low-rank quadratic strata, but do not classify arbitrary nonlinear hypersurfaces in the coefficient space.*

APPENDIX F. THE MINIMAL SMOOTH DEFECT: FORMAL RIGIDITY AND SIGN TORSORS

This appendix records what can be proved about the first possible nonflat cubic normalization. It does not exclude that defect: the final end-to-global step remains open.

Assume that the local defect has length one, that the discriminant has its minimal multiplicity six, and that the resulting leading plane cubic $C_\phi \subset \mathbb{P}^2$ is smooth. On the exceptional plane $E = \mathbb{P}^2$, let Q be the rank-two quotient bundle and put

$$W_k = H^0(E, \text{Sym}^3 Q(k)).$$

The order- k coordinate-and-frame gauge space is

$$A_k = H^0(E, \text{At}(\Omega_E^1)(k)).$$

For the leading cubic ϕ , the infinitesimal action is

$$\delta_{\phi,k}(A) = \left[\sum_{i,j} A_{ij}(x) q_i \frac{\partial \phi}{\partial q_j} \right] \in W_k.$$

Theorem F.1 (Formal normal rigidity). *If C_ϕ is smooth, then*

$$\delta_{\phi,k}: A_k \longrightarrow W_k$$

is surjective for every $k \geq 1$. Consequently every normalized formal datum

$$\eta = s^2(\phi + s\psi_1 + s^2\psi_2 + \cdots)$$

is formally gauge-equivalent, with its leading cubic fixed, to $s^2\phi$.

Proof. There is a natural bundle map

$$V^\vee \otimes Q \longrightarrow \mathrm{Sym}^3 Q.$$

On a line H , its image is the product of the binary linear forms with the span of the restrictions of the three partial derivatives of ϕ . Those binary quadratics have no common zero when C_ϕ is smooth, and therefore generate every binary cubic after multiplication by linear forms. The bundle map is consequently surjective.

If K_ϕ is its kernel, an exact Hesse-form calculation at twist one gives

$$\det \delta_{\phi_\lambda, 1} = -3^{24}(\lambda^3 - 1)^6.$$

The determinant is nonzero precisely on the smooth Hesse locus. Together with the Chern classes of K_ϕ , the resulting cohomology vanishing identifies $K_\phi \simeq \Omega_{\mathbb{P}^2}^1$. Thus

$$0 \longrightarrow \Omega_{\mathbb{P}^2}^1 \longrightarrow \mathrm{At}(\Omega_{\mathbb{P}^2}^1) \xrightarrow{\delta_\phi} \mathrm{Sym}^3 Q \longrightarrow 0.$$

After twisting, $H^1(\mathbb{P}^2, \Omega^1(k)) = 0$ for $k \geq 1$, which proves surjectivity. Removing ψ_k successively gives the formal normalization. The exact script `cubic_jet_rank_check.py`, stored under `code/minimal-smooth-defect/`, checks the Hesse determinant and the first Fermat ranks over $\mathbb{Q}$. $\square$

Remark F.2. At $k = 0$, the cokernel is $H^1(\mathbb{P}^2, \Omega^1) \simeq \mathbb{C}$, the expected modulus of a plane cubic. Thus the formal normal direction introduces no additional modulus. The statement is formal on the blown-up incidence model; compatibility with the unsaturated Rees lattice is part of any later descent through the blowdown.

We next pass to the S_3 -Galois closure. Its exceptional surface is

$$A = C_\phi \times C_\phi.$$

Using ordered collinear triples (p_1, p_2, p_3) , with $p_1 + p_2 + p_3 = 0$, the three transposition-fixed collision curves are

$$\Gamma_{12} = (t, t), \quad \Gamma_{23} = (-2t, t), \quad \Gamma_{31} = (t, -2t).$$

Theorem F.3 (Local sign-3 torsors). *The 3-torsion line bundles on A that restrict trivially to all three collision curves form*

$$K_{\mathrm{loc}} = \ker \left(\mathrm{Pic}^0(A)[3] \longrightarrow \prod_{i < j} \mathrm{Pic}^0(\Gamma_{ij})[3] \right) \simeq C_\phi[3] \simeq \mathbb{F}_3^2.$$

The subgroup $A_3 \subset S_3$ fixes K_{loc} , while a transposition acts by inversion. Each class is represented by an algebraic finite étale cover of the proper henselian local Galois end.

Proof. Write a degree-zero line bundle as

$$L_{\alpha, \beta} = \mathrm{pr}_1^* P_\alpha \otimes \mathrm{pr}_2^* P_\beta.$$

Its three restrictions correspond respectively to

$$\alpha + \beta, \quad -2\alpha + \beta, \quad \alpha - 2\beta.$$

They vanish simultaneously exactly when $\beta = -\alpha$ and $3\alpha = 0$, proving the first assertion. A three-cycle fixes $L_{\alpha, -\alpha}$, whereas a transposition sends it to its inverse.

For nonzero α , the algebra

$$\mathcal{O}_A \oplus L_\alpha \oplus L_\alpha^{\otimes 2}$$

is a connected finite étale μ_3 -torsor. Finite étale covers of a proper scheme over a henselian local ring are equivalent to finite étale covers of its closed fiber, so these torsors extend uniquely over the henselian local Galois completion. $\square$

Question F.4 (End-to-global sign torsor). *Does at least one nonzero class of K_{loc} extend across the global Keller boundary with the required sign equivariance and split transposition inertia? Such a class would descend to a nontrivial finite étale degree-three cover of $\mathbb{A}^3$, which is impossible.*

The omitted value itself gives a useful global fibration.

Proposition F.5 (Direction fibration from an omitted value). *Let $F: \mathbb{A}^3 \rightarrow \mathbb{A}^3$ be a complex Keller map and $p \notin F(\mathbb{A}^3)$. Then*

$$g_p: \mathbb{A}^3 \longrightarrow \mathbb{P}^2, \quad x \longmapsto [F(x) - p]$$

is smooth of relative dimension one, and its image is the complement of at most finitely many points.

Proof. The differential of projectivization is the quotient by the line $\langle F(x) - p \rangle$. Composing it with the invertible matrix DF_x has rank two, so g_p is smooth. Its image is open. If its complement contained a curve $V(H)$, the polynomial $H(F - p)$ would be nowhere zero on affine space and hence constant. The injectivity of $F^*: \mathbb{C}[a, b, c] \rightarrow \mathbb{C}[x, y, z]$ would then make the nonconstant homogeneous polynomial $H(a - p)$ constant, a contradiction. $\square$

The remaining obstruction in question F.4 is therefore carried by the horizontal boundary of a smooth curve fibration over a cofinite open subset of $\mathbb{P}^2$, not by complicated base monodromy.

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checks are identified in section 7; the remaining arguments require ordinary mathematical review.

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